MIL-I-15349B
18 November 1963
SUPERSEDING
MIL-I-15349A
25 August 1952

MILITARY SPECIFICATION

INSULATION TAPE, THERMAL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. - This specification covers thermal insulating tape for use as an insulation for piping and tubing at temperatures up to 750° F. It is especially suitable for small piping and tubing where space conditions render awkward the use of molded insulation covering.

1.2 Classification - Thermal insulating tape shall be of the following types, as specified (see 6.1):

Type I-For spiral wrapping.
Type II-For lateral wrapping.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
UU-P-268 - Paper, Kraft, Untreated, Wrapping.
CCC-T-191 - Textile Test Methods.
PPP-B-636 - Box, Fiberboard.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129 - Marking For Shipment And Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS
ASTM-D1918 - Asbestos Content of Asbestos Textiles, Method of Test for.

FSC 5640

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS.

3.1 Material. - Thermal insulating tape shall be composed of a woven asbestos (chrysotile) jacket enclosing an asbestos fiber or fibrous glass felting or sliver.

3.2 Jacket. -

3.2.1 Construction. - The jacket for the hot side and not less than 1/4 inch on the edges of the finished tape shall be woven from asbestos yarn containing not less than 90 percent by weight of chrysotile asbestos fiber. The back or cold side of the tape shall be woven from asbestos yarn containing not less than 80 percent by weight of chrysotile asbestos fiber. The jacket may be either tubular woven in one piece or fabricated of asbestos cloth sewed into tubular form.

3.2.1.1 Weave. - The jacket shall be plain weave and consist of not less than 16 strands per inch in the warp and not less than 7 strands per inch in the filling. (The jacket may have an open weave on the hot side consisting of not less than 8 strands per inch in the warp and 7 strands per inch in the filling) (see 4.3.2).

3.2.2 Breaking strength. - The breaking strenght of samples of the asbestos cloth jacket from the cold side of the tape shall be not less than 60 pounds per inch width (see 4.3.4).

3.3 Felting or sliver. -

3.3.1 Construction. - The felting or sliver may be composed of one or more fabricated or felted rovings of either fibrous glass or carded amosite asbestos fibers containing not less than 95 percent be weight, of asbestos. When a single felted roving construction is used, the felting or sliver shall be suitably reinforced and held in shape as a single unit by 8 or more loosely spiralled or braided 2-ply asbestos or glass-fiber strands. When a fabrication of more than one fibrous glass or asbestos-fiber roving is used to construct the felting or sliver, the rovings shall be held in place and together by a one-over-one-under-plain weave or not less than 3 warp ends twill weave of 2-ply glass or asbestos fiber yarns spaced approximately 7/8 inch apart.

3.3.1.1 Reinforcing strands. - Plain and wire-inserted reinforcing strands will be allowed, if required, in the construction.

3.3.1.2 Wire. - The wire used in reinforcing strands shall be of brass, copper, or nickel-copper alloy, 0.008 inch in diameter, plus or minus 0.001 inch.

3.3.1.3 Yarn. - The asbestos or glass-fiber yarns used in the supporting structure shall be two-ply. The asbestos-fiber yarn shall contain not less than 90 percent, by weight, of asbestos fiber.

3.3.1.4 Fineness of fibers. - The diameter of individual glass fibers shall average not more than 0.0008 inch. The maximum diameter of any fiber in the test sample shall be not more than 0.0015 inch (see 4.3.5).

3.3.2 Alkalinity. - The alkalinity of the fibrous glass, expressed as sodium oxide, shall not exceed 0.60 percent (see 4.3.6).

3.4 Loss in weight. - Tape in finished form shall not show a loss in weight greater than 15 percent, after subjection to the test specified in 4.3.7.

3.5 Type I-for spiral wrapping. -

3.5.1 Weave. - The open weave of the jacket on the hot side shall be not more than 90 percent of the nominal width of the material.

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3.5.2 Dimensions and weight. - Unless otherwise specified, thermal insulating tape shall be furnished as specified in table I (see 6.1).

Table I - Dimensions and weight

Width nominal	Thickness nominal	Weight nominal
inch ± 1/16	inch ± 1/16	lbs. ± 0.5/25 ft. roll
1-1/2	1/4	2.5
2	1/4	4.0
3	1/4	5.0
2-3/8	3/8	4.0
2-1/2	5/8	4.5

3.5.3 Jacket weight. - The tubular jacket shall be not more than 70 percent, by weight, of the finished tape made with either a fibrous glass or an asbestos fiber felting or sliver.

3.6 Type II. - for lateral wrapping.

3.6.1 Weave. - The open weave on the hot side of the jacket shall be not more than 70 percent of the nominal width of the material.

3.6.2 Dimensions and weight. - Unless otherwise specified, thermal insulating tape shall be as specified in table II (see 6.1):

Table II - Dimensions and weight

Width	Thickness	Weight
inch ± 1/16	inch ± 1/16	lbs. ± 0.5/25 ft. roll
4	3/8	8.5
5-1/4	3/8	10.5
5-1/2	3/8	11.0
5-3/4	3/8	11.5
7-1/4	3/8	11.25
8-1/4	3/8	13.25
4	3/4	7.75
5	3/4	9.00
5-1/4	3/4	9.50
5-1/2	3/4	10.00
5-3/4	3/4	10.50
6-1/2	3/4	11.75
6-7/8	3/4	12.50
5-1/2	7/8	11.25
4	1	9.25
5	1	11.50
6-7/8	1	15.50
8	1	18.00

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3.6.3 Jacket weight. - The tubular jacket shall be not more than 55 percent, by weight, of finished tape made with either a fibrous glass or an asbestos fiber felting or sliver.

3.7 Marking. - The tape shall be stamped or marked in a legible manner so as to identify either the "hot side", or "cold side".

3.8 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory satisfactory to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection. - Sampling for quality conformance inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter. For pupposes of sampling, an inspection lot for examination and tests shall consist of all tape of the same type offered for delivery at one time.

4.2.1 Sampling for examination. - A random sample of rolls shall be selected from each lot of material for examination specified in 4.2.3.1 with lot acceptance based on Inspection Level I, and an Acceptable Quality Level of 4.0 percent defective.

4.2.2 Sampling for tests. - From each lot, three samples, 1 foot long, shall be selected at random for the tests specified in 4.3. Each specimen shall be taken from a different roll.

4.2.3 Examination and tests. -

4.2.3.1 Examination. - Each of the sample rolls selected in accordance with 4.2.1 shall be visually and dimensionally examined to determine conformance with the requirements of this specification which do not require tests. Examination shall be conducted as specified in table III. Any roll in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective rolls in any sample exceeds the acceptance number for that sample, shall be cause for rejection of the lot represented by the sample.

Table III - Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	None defined.
Major:	
101	Type of insulation not as specified.
102	Material less than minimum requirements, evidence of unauthorized material; insulation not composed of specified materials.
103	Reinforcing wire not brass, copper, or nickel-copper alloy.
104	Diameter of reinforcing wire not within tolerance allowed.
105	Construction non-conforming or not as specified.
106	Dimensions, thickness and width not as specified.
107	Evidence of inferior workmanship.
Minor	None defined.

4.2.3.2 Testing. - The sample selected in accordance with 4.2.2 shall be subjected to the tests specified in 4.3 to determine conformance with this specification.

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4.2.3.2.1 Action in case of failure. - If any sample tested is found to be not in conformance with this specification, this shall be cause for rejection of the lot which it represents.

4.3 Test procedures. -

4.3.1 Asbestos content. - The asbestos content shall be determined in accordance with method specified in ASTM D1918.

4.3.2 Jacket construction. - The number of warp yarns and the number of filling yarns shall be determined by method 5050 of CCC-T-191.

4.3.3 Width. - The width of the tape shall be determined when it is compressed to the specified nominal thickness.

4.3.4 Breaking strength. - The breaking strength of the asbestos cloth jacket shall be determined by method 5104 of CCC-T-191.

4.3.5 Diameter of fiber. - Diameter of fibers shall be determined microscopically on the basis of measuring seven fibers on each of the samples selected in accordance with 4.2.2. The average diameter for purposes of determining conformance with 3.3.1.4 shall be the average of all measurements on all samples.

4.3.6 Alkalinity. - The alkalinity test shall be performed as follows: Weigh a 5-gram (±0.01 gram) representative sample of the fibrous glass, and introduce into a 500-milliliter (mℓ.), pyrex Erlenmeyer flask. Wet with 5 mℓ. of 95 percent ethyl alcohol and add 400 mℓ. of distilled water. Reflux for 4 hours ±5 minutes. At the end of this period, disconnect the condenser and filter at once through No. 41 Whatman paper supported in a Buchners funnel. Wash the flask and material three times with 25-mℓ. portions of hot distilled water using suction. Titrate immediately with 0.02N H_2SO_4, using six to eight drops of 1 percent solution of phenol-red indicator, to the disappearance of the pink color. Run a blank determination on the same amount of distilled water and alcohol and correct for any alkalinity shown. The percentage alkalinity as Na_2O shall be calculated from the following formula:

$$\text{Percent } Na_2O = \frac{(A-B)\ N \times 0.031 \times 100}{W}$$

where A = ml. H_2SO_4 to titrate sample
B = ml. H_2SO_4 to titrate blank
N = normality of the H_2SO_4
W = weight of sample grams

4.3.7 Loss in weight. - Compliance with 3.4 shall be determined by subjecting not less than 1 linear foot to soaking heat for 6 hours at approximately 750° F.; the weight before and after heating shall be determined.

$$\text{Loss in weight, percent} = \frac{\text{Weight before heating-weight after heating X 100}}{\text{Weight before heating}}$$

4.3.8 Weight. - The tape shall be laid out flat. A specimen of not less than 3 linear feet shall be cut from each roll and weighed to the nearest 0.1 ounce. For type I tape, calculate the weight to pounds per inch of width per 100 linear feet using width as determined by the method specified in 4.3.3. For type II tape, calculate the weight to pounds per 100 linear feet. In both cases remove the jacket, weight to the nearest 0.1 ounce and calculate the percent by weight of jacket in the finished tape to verify conformance to 3.6.3.

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4.3.8.1 Weigh each of the sample rolls selected for 4.2.1 to verify conformance to 3.6.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Level A. - Tape in widths as specified (see 6.1) shall be packaged in rolls 25 linear feet in length. Each roll shall be completely wrapped in kraft paper conforming to UU-P-268.

5.1.2 Level C. - Packaging shall be sufficient to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity for immediate use. This may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. - Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. - Rolls, packaged as specified (see 6.1) shall be packed in containers conforming to Class 2 of PPP-B-636. All center and edge seams and manufacturer's joints shall be sealed and waterproofed with pressure sensitive tape in accordance with the appendix to PPP-B-636.

5.2.2 Level B. - Rolls, packaged as specified (see 6.1), shall be packed in containers conforming to Class 1 of PPP-B-636, and closed as specified therein.

5.2.3 Level C. - Rolls, packaged as specified (see 6.1), shall be packed in containers, at the lowest rates, in a manner which will insure acceptance by common carrier and will afford protection against physical damage during direct shipment from the supply source to the first receiving activity for immediate use. This level in general shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Width and thickness required (see 3.5.2 and 3.6.2).
(d) Levels of packaging and packing required (see 5.1 and 5.2).
(e) Special marking if required (see 5.3).

Custodians:
Army - MO
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Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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